Network Management and Automation

Lab 6

DevOps - Ansible

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Network Engineering Program

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Summary

Ansible has proved to be profoundly different from other configuration management tools available today, and is prevalent in industry for managing and configuring network devices as well as servers. It has been designed to make configurations easy in almost every way. Some of its notable features include its simple English configuration syntax, agentless running service, and its ease of setup.

Objectives

- Configuring and installing Ansible.
- Learn to use Playbooks for configuration management.
- Learn how to perform package management on the hosts.
- Learn how to perform identity and access management.
- Learn to perform file transfers and directory management.

Problem Statement

Liverpool and Real Madrid have a strong history of European Football. The two teams have once again made their way to an epic Champions League final. The FIFA administration has been entrusted the job to ensure that all things are in place to give us a thriller.

You will be acting as the FIFA administrator and be performing the tasks given below.

Objective 1 – Setting up a Virtual Network

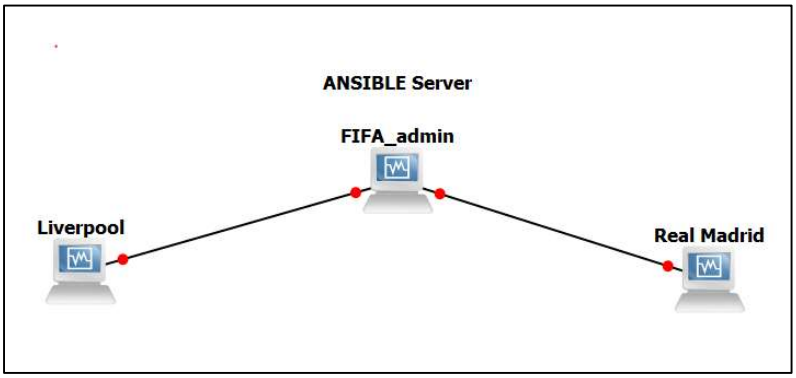

Figure 1

1. Create the above network of three virtual machines (use the NetMan VM as the FIFA_admin, which will act as the Ansible Server) in VMWare Workstation.
   We have assigned 2 additional VMs in Vsphere for you to use. You should be able to see these VMs once you login to vsphere

   OR

   You can use any Linux based OS of your choice to spawn the Liverpool and Real Madrid virtual machines. You can use the Ubuntu 16.04 Desktop image as the OS (https://releases.ubuntu.com/16.04.7/). Liverpool Internal Network- liverpool-192.168.10.1 and netman-192.168.10.2. Realmadrid Internal Network - realmadrid-172.16.10.1 and netman-172.16.10.2

2. Provide screenshots of successful network connectivity (ping between netman and Liverpool, and netman and realmadrid). **[10 Points]**

```
netman@netman-VirtualBox:~$ ping Liverpool
PING Liverpool (10.224.78.169) 56(84) bytes of data.
64 bytes from Liverpool (10.224.78.169): icmp_seq=1 ttl=64 time=0.371 ms
64 bytes from Liverpool (10.224.78.169): icmp_seq=2 ttl=64 time=0.256 ms
64 bytes from Liverpool (10.224.78.169): icmp_seq=3 ttl=64 time=0.290 ms
64 bytes from Liverpool (10.224.78.169): icmp_seq=4 ttl=64 time=0.247 ms
^C
--- Liverpool ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3097ms
rtt min/avg/max/mdev = 0.247/0.291/0.371/0.048 ms
netman@netman-VirtualBox:~$ ping RealMadrid
PING RealMadrid (10.224.76.65) 56(84) bytes of data.
64 bytes from RealMadrid (10.224.76.65): icmp_seq=1 ttl=64 time=0.335 ms
64 bytes from RealMadrid (10.224.76.65): icmp_seq=2 ttl=64 time=0.241 ms
64 bytes from RealMadrid (10.224.76.65): icmp_seq=3 ttl=64 time=0.240 ms
64 bytes from RealMadrid (10.224.76.65): icmp_seq=4 ttl=64 time=0.251 ms
^C
--- RealMadrid ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3108ms
rtt min/avg/max/mdev = 0.240/0.266/0.335/0.039 ms
```

Objective 2 – Installing Ansible

1. Issue the "**ansible --version"** command to check if ansible is present in FIFA_admin. If not, install it using pip3. Provide screenshot of the output. **[5 Points]**

```
netman@netman-VirtualBox:~$ ansible --version
ansible [core 2.14.4]
  config file = None
  configured module search path = ['/home/netman/.ansible/plugins/modules', '/usr/share/ansible/plugins/modules']
  ansible python module location = /usr/local/lib/python3.10/dist-packages/ansible
  ansible collection location = /home/netman/.ansible/collections:/usr/share/ansible/collections
  executable location = /usr/local/bin/ansible
  python version = 3.10.12 (main, Jan 26 2026, 14:55:28) [GCC 11.4.0] (/usr/bin/python3)
  jinja version = 3.0.3
  libyaml = True
```

2. In order to allow FIFA_admin to control the two hosts through a playbook, ensure that you add the hosts in the /etc/ansible/hosts or any other Inventory file of your choice (https://docs.ansible.com/projects/ansible/latest/inventory_guide/index.html). Ensure connectivity between the server and managed hosts using the "**ansible all -m ping --ask-pass**" command. Provide screenshot of the output. **[5 Points]**

```
netman@netman-VirtualBox:~/Documents/python_code/lab6$ ansible all -m ping -u student --ask-pass -i
 ansible-inventory.ini
SSH password:
RealMadrid | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
Liverpool | SUCCESS => {
    "ansible_facts": {
        "discovered_interpreter_python": "/usr/bin/python3"
    },
    "changed": false,
    "ping": "pong"
}
```

Objective 3 – Creating a Centralized Playbook

The following objectives need to be accomplished by creating a **single** playbook on the FIFA_admin machine. The playbook file should be named as "YourIdentiKey_play.yml". Make sure you give meaningful names to the tasks performing the below objectives in the playbook.

Objective 3.1 – Installing Packages

1. Liverpool and Real Madrid machines should have FTP and HTTP capabilities. Additionally they need pip and expect packages. Install all these essential packages through the playbook on FIFA_admin (use loops to perform the repetitive installation tasks).
   [10 Points]

   References for Linux Packages to allow the above capabilities
   https://linuxvox.com/blog/install-ftp-linux/
   https://linuxvox.com/blog/http-a-ubuntu/
   https://www.geeksforgeeks.org/installation-guide/how-to-install-pip-in-ubuntu/
   https://ioflood.com/blog/install-expect-command-linux/

   https://www.redhat.com/en/blog/software-packages-ansible
   https://docs.ansible.com/projects/ansible/latest/playbook_guide/playbooks_privilege_escalation.html#using-become

```
- name: My first play
  hosts: clubhosts
  tasks:
    - name: Ping Hosts
      ansible.builtin.ping:

    - name: Print Message
      ansible.builtin.debug:
        msg: Hello Champions League

    - name: install pip
      ansible.builtin.apt:
        name: "{{ item }}"
        state: latest
      become: true
      loop:
        - pip
        - expect
        - apache2
        - vsftpd
```

2. Provide screenshots of the relevant portion of the playbook and the successful package installation on the two nodes. **[10 Points]**

```
TASK [install pip] ************************
ok: [Liverpool] => (item=pip)
ok: [RealMadrid] => (item=pip)
changed: [RealMadrid] => (item=expect)
changed: [Liverpool] => (item=expect)
changed: [RealMadrid] => (item=apache2)
changed: [Liverpool] => (item=apache2)
changed: [RealMadrid] => (item=vsftpd)
changed: [Liverpool] => (item=vsftpd)
```

```
student@csci5180-vm2-zaco6003:~$ expect -version
expect version 5.45.4
student@csci5180-vm2-zaco6003:~$ apache2 -version
Server version: Apache/2.4.58 (Ubuntu)
Server built:   2025-12-09T15:50:28
student@csci5180-vm2-zaco6003:~$ vsftpd -version
vsftpd: version 3.0.5
student@csci5180-vm2-zaco6003:~$
```

```
student@csci5180-vm3-zaco6003:~$ pip --version
pip 24.0 from /usr/lib/python3/dist-packages/pip (python 3.12)
student@csci5180-vm3-zaco6003:~$ expect -version
expect version 5.45.4
student@csci5180-vm3-zaco6003:~$ apache2 -version
Server version: Apache/2.4.58 (Ubuntu)
Server built:   2025-12-09T15:50:28
student@csci5180-vm3-zaco6003:~$ vsftpd -version
vsftpd: version 3.0.5
student@csci5180-vm3-zaco6003:~$
```

Objective 3.2 – Creating Users, Groups, and Permissions

1. Create users and groups on the managed hosts using the same playbook on the control machine. **[5 Points]**

```
TASK [configure defenders] ************************************
ok: [Liverpool] => (item=jcarragher)
ok: [Liverpool] => (item=dagger)

TASK [configure captain] **************************************
ok: [Liverpool]

PLAY [Configure Real Madrid Players] ****************************

TASK [Gathering Facts] ****************************************
ok: [RealMadrid]

TASK [configure attackers] ************************************
changed: [RealMadrid] => (item=lfigo)
changed: [RealMadrid] => (item=rsoldado)

TASK [configure defenders] ************************************
changed: [RealMadrid] => (item=rcarlos)
changed: [RealMadrid] => (item=fcannavaro)

TASK [configure captain] **************************************
changed: [RealMadrid]

PLAY RECAP ****************************************************
Liverpool                  : ok=9    changed=0    unreachable=0
RealMadrid                 : ok=9    changed=3    unreachable=0
```

```
hkewell:x:1003:1006::/home/hkewell:/bin/sh
mowen:x:1004:1007::/home/mowen:/bin/sh
jcarragher:x:1005:1008::/home/jcarragher:/bin/sh
dagger:x:1006:1009::/home/dagger:/bin/sh
sgerrard:x:1007:1010::/home/sgerrard:/bin/sh
```

```
lfigo:x:1003:1006::/home/lfigo:/bin/sh
rsoldado:x:1004:1007::/home/rsoldado:/bin/sh
rcarlos:x:1005:1008::/home/rcarlos:/bin/sh
fcannavaro:x:1006:1009::/home/fcannavaro:/bin/sh
zzidane:x:1007:1010::/home/zzidane:/bin/sh
```

2. The three groups on each machine are: Attack, Defense, and Captain. The users to be created, their corresponding groups, and permissions are mentioned in tables below. The users should be created with the naming format of FirstInitialLastName (for example Harry Kewell's user name would be hkewell). **[15 Points]**

https://oneuptime.com/blog/post/2026-02-21-how-to-define-host-variables-in-ansible-inventory/view

```
attacker1: hkewell
attacker2: mowen
defender1: jcarragher
defender2: dagger
captain: sgerrard
```

```
attacker1: lfigo
attacker2: rsoldado
defender1: rcarlos
defender2: fcannavaro
captain: zzidane
```

- **Name: add groups**
  Ansible.builtin.group:
    Name: “{{ item }}”
    State: present
  Become: true
  Loop:
    - **Attack**
    - **Defense**
    - **Captain**

```
- name: add attackers
  ansible.builtin.user:
    name: "{{ item }}"
    groups: Attack
    append: true
    umask: 0111
  become: true
  loop:
    - "{{ attacker1 }}"
    - "{{ attacker2 }}"

- name: configure defenders
  ansible.builtin.user:
    name: "{{ item }}"
    groups: Defense
    append: true
    umask: 0666
  become: true
  loop:
    - "{{ defender1 }}"
    - "{{ defender2 }}"

- name: configure captain
  ansible.builtin.user:
    name: "{{ captain }}"
    groups: Captain, sudo
    append: true
    umask: 0000
  become: true
```

Liverpool

User	Group/Groups	Group Permissions
Harry Kewell		rw-rw-rw

Michael Owen	Attack	
Jamie Carragher	Defense	--x--x--x
Daniel Agger		
Steven Gerrard (present in two groups)	Captain and Any group with highest access permissions (Eg: Wheel)	rwxrwxrwx

Real Madrid

User	Group/Groups	Group Permissions
Luis Figo	Attack	rw-rw-rw
Roberto Soldado		
Roberto Carlos	Defense	--x--x--x
Fabio Cannavaro		
Zinedine Zidane (present in two groups)	Captain and Any group with highest access permissions (Eg: Wheel)	rwxrwxrwx

3. Provide screenshots of the relevant portion of the playbook, successful creation of all users and their group permissions. **[5 Points]**

```yaml
- name: add attackers
  ansible.builtin.user:
    name: "{{ item }}"
    groups: Attack
    append: true
    umask: 0111
  become: true
  loop:
    - "{{ attacker1 }}"
    - "{{ attacker2 }}"

- name: configure defenders
  ansible.builtin.user:
    name: "{{ item }}"
    groups: Defense
    append: true
    umask: 0666
  become: true
  loop:
    - "{{ defender1 }}"
    - "{{ defender2 }}"

- name: configure captain
  ansible.builtin.user:
    name: "{{ captain }}"
    groups: Captain, sudo
    append: true
    umask: 0000
  become: true
```

```
sudo:x:27:vmware,student,instructor,sgerrard
audio:x:29:
dip:x:30:vmware
```

```
sudo:x:27:vmware,student,instructor,zzidane
audio:x:29:
dip:x:30:vmware
```

```
Attack:x:1003:hkewell,mowen
Defense:x:1004:jcarragher,dagger
Captain:x:1005:sgerrard
hkewell:x:1006:
mowen:x:1007:
jcarragher:x:1008:
dagger:x:1009:
sgerrard:x:1010:
```

```
Attack:x:1003:lfigo,rsoldado
Defense:x:1004:rcarlos,fcannavaro
Captain:x:1005:zzidane
lfigo:x:1006:
rsoldado:x:1007:
rcarlos:x:1008:
fcannavaro:x:1009:
zzidane:x:1010:
```

Objective 3.3 – Creating Files and Directories

1. Add commands to the same playbook to create directories on Liverpool and Real Madrid which will be used to store route table information in the next objective. The directory

should be created in the path /MachineName_Admin/Route_Info/. The owner and group owner for this directory should be Captain, and they should have all permissions. Users who are not in the Captain group should have read-only access to this directory.
Ex: LiverPool_Admin/Route_Info rwx/rwx/r—
https://spacelift.io/blog/ansible-create-directory

- Name: configure captain directory
  Become: true
  Ansible.builtin.file:
    Path: "/{{ inventory_hostname }}_ Admin/Route_Info/"
    State: directory
    Mode: "774"
    Owner: "{{ captain }}"
    Group: Captain

2. Provide screenshots of the relevant portion of the playbook, successful creation of the directory and applied permissions. [**10 Points**]

```
TASK [configure captain] ***********************************************************************************
ok: [RealMadrid]
ok: [Liverpool]

TASK [create captain directory] ****************************************************************************
changed: [RealMadrid]
changed: [Liverpool]

PLAY RECAP *************************************************************************************************
Liverpool                  : ok=9    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
RealMadrid                 : ok=9    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
drwxrwxr--   3 zzidane Captain       4096 Mar  1 19:52 RealMadrid_Admin
drwx------   7 root    root          4096 Mar  1 16:48 root
drwxr-xr-x  47 root    root          1420 Mar  1 06:30 run
lrwxrwxrwx   1 root    root             8 Apr 22  2024 sbin -> usr/sbin
drwxr-xr-x   2 root    root          4096 Mar 31  2024 sbin.usr-is-merged
```

```
drwxrwxr--   3 sgerrard Captain       4096 Mar  1 19:52 Liverpool_Admin
drwx------   2 root     root         16384 Aug 31 19:37 lost+found
drwxr-xr-x   2 root     root          4096 Feb 15  2025 media
drwxr-xr-x   2 root     root          4096 Feb 15  2025 mnt
drwxr-xr-x   4 root     root          4096 Sep  1 18:02 opt
```

```yaml
- name: configure captain
  ansible.builtin.user:
    name: "{{ captain }}"
    groups: Captain, sudo
    append: true
    umask: 0000
  become: true

- name: create captain directory
  become: true
  ansible.builtin.file:
    path: "/{{ inventory_hostname }}_Admin/Route_Info/"
    state: directory
    mode: "774"
    owner: "{{ captain }}"
    group: Captain
```

3. On each host, create a file Captain_Details.txt under the above created directory and store the captain information (full name, nationality, username) using the same playbook. Provide relevant screenshots. [**10 Points**]
   https://www.ansiblepilot.com/articles/write-a-variable-to-a-file-ansible-module-copy-vs-template

```
TASK [create captain details] ***********************************************************************************************
changed: [Liverpool] => {"changed": true, "dest": "/Liverpool_Admin/Route_Info/Captain_Details.txt", "gid": 1005, "group": "Captain"
, "mode": "0774", "owner": "sgerrard", "size": 35, "state": "file", "uid": 1007}
changed: [RealMadrid] => {"changed": true, "dest": "/RealMadrid_Admin/Route_Info/Captain_Details.txt", "gid": 1005, "group": "Captai
n", "mode": "0774", "owner": "zzidane", "size": 35, "state": "file", "uid": 1007}

TASK [insert name into file] ************************************************************************************************
changed: [RealMadrid] => {"changed": true, "checksum": "21db502fed810b2badb20c8c13dc510ca6599a2f", "dest": "/RealMadrid_Admin/Route_
Info/Captain_Details.txt", "gid": 1005, "group": "Captain", "md5sum": "34c178e58ef6f2d84d733e390b8acfd9", "mode": "0774", "owner": "
zzidane", "size": 31, "src": "/home/student/.ansible/tmp/ansible-tmp-1772423315.2382736-38589-86854413253915/source", "state": "file
", "uid": 1007}
changed: [Liverpool] => {"changed": true, "checksum": "31e93203aac5122c840705463893aa3b245cbdb4", "dest": "/Liverpool_Admin/Route_In
fo/Captain_Details.txt", "gid": 1005, "group": "Captain", "md5sum": "62e75ed616f2773bbecc68038586184d", "mode": "0774", "owner": "sg
errard", "size": 31, "src": "/home/student/.ansible/tmp/ansible-tmp-1772423315.2538345-38590-153832597933339/source", "state": "file
", "uid": 1007}
```

```
- name: create captain directory
  become: true
  ansible.builtin.file:
    path: "/{{ inventory_hostname }}_Admin/Route_Info/"
    state: directory
    mode: "774"
    owner: "{{ captain }}"
    group: Captain

- name: create captain details
  become: true
  ansible.builtin.file:
    path: "/{{ inventory_hostname }}_Admin/Route_Info/Captain_Details.txt"
    state: touch
    mode: "774"
    owner: "{{ captain }}"
    group: Captain

- name: insert name into file
  become: true
  ansible.builtin.copy:
    content: "{{ captain_fullname }}\n{{ captain_nationality }}\n{{ captain }}\n"
    dest: "/{{ inventory_hostname }}_Admin/Route_Info/Captain_Details.txt"
```

```
student@csci5180-vm2-zaco6003:~$ sudo cat /RealMadrid_Admin/Route_Info/Captain_Details.txt
Zinedine Zidane
French
zzidane
```

```
student@csci5180-vm3-zaco6003:~$ sudo cat /Liverpool_Admin/Route_Info/Captain_Details.txt
Steven Girard
English
sgerrard
```

Objective 3.4 – Executing Shell Commands, Performing File Transfer

1. The FIFA_admin machine needs to request route table and top ten running processes (based on memory usage) from the managed hosts. Add functionality in the playbook to do the following:

a. Execute shell commands (hint: netsat , ps) to retrieve the required information and store it in two different files. **[10 Points]**
https://linuxize.com/post/ps-command-in-linux/
https://vitux.com/how-to-view-the-network-routing-table-in-ubuntu/

Commands:

Netstat -rn

Ps aux –sort=-%mem | head -n 11

```
- name: create shell_output
  become: true
  ansible.builtin.file:
    path: "/{{ inventory_hostname }}_Admin/Route_Info/shell_output.txt"
    state: touch
    mode: "775"
    owner: "{{ captain }}"
    group: Captain

- name: get routing table
  become: true
  ansible.builtin.shell:
    cmd: "netstat -rn > /{{ inventory_hostname }}_Admin/Route_Info/shell_output.txt"

- name: get top talkers
  become: true
  ansible.builtin.shell:
    cmd: "ps aux --sort=-%mem | head -n 11 >> /{{ inventory_hostname }}_Admin/Route_Info/shell_output.txt"
```

```
student@csci5180-vm3-zaco6003:~$ sudo cat /Liverpool_Admin/Route_Info/shell_output.txt
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
0.0.0.0         10.224.76.1     0.0.0.0         UG        0 0          0 ens33
10.224.76.0     0.0.0.0         255.255.252.0   U         0 0          0 ens33
172.17.0.0      0.0.0.0         255.255.0.0     U         0 0          0 docker0
192.168.122.0   0.0.0.0         255.255.255.0   U         0 0          0 virbr0
USER         PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
jenkins     1589  0.0  4.6 4698004 378640 ?      Ssl  Jan09  71:06 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/je
broot=/var/cache/jenkins/war --httpPort=8080
root        2958  0.4  4.1 1983160 333720 ?      Sl   Jan09 300:21 falcon-sensor-bpf
student   833606  0.0  3.5 3891612 284716 ?      Ssl  Feb24   2:14 /usr/bin/gnome-shell
student   834096  0.0  1.1 275216 90528 ?        S    Feb24   0:00 /usr/bin/Xwayland :0 -rootless -noreset -accessx -core -auth /
.mutter-Xwaylandauth.Q0RZK3 -listenfd 4 -listenfd 5 -displayfd 6 -initfd 7 -byteswappedclients
student   834419  0.0  1.1 738064 90312 ?        Sl   Feb24   0:00 /usr/libexec/mutter-x11-frames
student   834305  0.0  1.0 657212 83288 ?        Ssl  Feb24   0:00 /usr/libexec/gsd-xsettings
root        1849  0.0  0.9 1983104 78892 ?       Ssl  Jan09   4:46 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/contain
student   833801  0.0  0.7 826372 63652 ?        Sl   Feb24   0:00 /usr/libexec/evolution-data-server/evolution-alarm-notify
root         355  0.0  0.7 112708 58720 ?        S<s  Jan09   0:38 /usr/lib/systemd/systemd-journald
student   834352  0.0  0.7 3068616 56900 ?       Sl   Feb24   1:05 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com/app/
/usr/share/gnome-shell/extensions/ding@rastersoft.com/app
```

b. Perform secure transfer of these files from host to FIFA_admin using **SCP** commands. Use the **expect** module to enter password and any other information associated with SCP when prompted. Passwords and prompts should not be handled manually but through the playbook. **[15 Points]**

Cat

```
TASK [scp output file] ***********************************************************************************************
changed: [RealMadrid] => {"changed": true, "cmd": "sudo scp /RealMadrid_Admin/Route_Info/shell_output.txt netman@10.224.77.6
:/home/netman/Documents/python_code/lab6/RealMadrid_output", "delta": "0:00:00.526833", "end": "2026-03-01 21:45:13.000235",
 "rc": 0, "start": "2026-03-01 21:45:12.473402", "stdout": "\rnetman@10.224.77.6's password: \r\n\rshell_output.txt
                    0%    0     0.0KB/s   --:-- ETA\rshell_output.txt                                    100% 1861     4.5MB/s
  00:00    ", "stdout_lines": ["", "netman@10.224.77.6's password: ", "", "shell_output.txt                                    0
%    0     0.0KB/s   --:-- ETA", "shell_output.txt                                    100% 1861     4.5MB/s   00:00    "]}
changed: [Liverpool] => {"changed": true, "cmd": "sudo scp /Liverpool_Admin/Route_Info/shell_output.txt netman@10.224.77.6:/
home/netman/Documents/python_code/lab6/Liverpool_output", "delta": "0:00:00.539611", "end": "2026-03-01 21:45:13.014134", "r
c": 0, "start": "2026-03-01 21:45:12.474523", "stdout": "\rnetman@10.224.77.6's password: \r\n\rshell_output.txt
                    0%    0     0.0KB/s   --:-- ETA\rshell_output.txt                                    100% 1853     2.4MB/s   0
0:00    ", "stdout_lines": ["", "netman@10.224.77.6's password: ", "", "shell_output.txt                                    0%
  0     0.0KB/s   --:-- ETA", "shell_output.txt                                    100% 1853     2.4MB/s   00:00    "]}

PLAY RECAP ***********************************************************************************************************
Liverpool                  : ok=15   changed=5    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
RealMadrid                 : ok=15   changed=5    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

```
netman@netman-VirtualBox:~/Documents/python_code/lab6$ cat Liverpool_output
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
0.0.0.0         10.224.76.1     0.0.0.0         UG        0 0          0 ens33
10.224.76.0     0.0.0.0         255.255.252.0   U         0 0          0 ens33
172.17.0.0      0.0.0.0         255.255.0.0     U         0 0          0 docker0
192.168.122.0   0.0.0.0         255.255.255.0   U         0 0          0 virbr0
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
jenkins      1589  0.0  4.6 4698004 378640 ?      Ssl  Jan09  71:07 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/ja
va/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080
root         2958  0.4  4.1 1983160 331788 ?      Sl   Jan09 300:55 falcon-sensor-bpf
student    833606  0.0  3.5 3891612 284748 ?      Ssl  Feb24   2:14 /usr/bin/gnome-shell
student    834096  0.0  1.1 275216 90528 ?        S    Feb24   0:00 /usr/bin/Xwayland :0 -rootless -noreset -accessx -core -a
uth /run/user/1001/.mutter-Xwaylandauth.Q0RZK3 -listenfd 4 -listenfd 5 -displayfd 6 -initfd 7 -byteswappedclients
student    834419  0.0  1.1 738064 90312 ?        Sl   Feb24   0:00 /usr/libexec/mutter-x11-frames
student    834305  0.0  1.0 657212 83296 ?        Ssl  Feb24   0:00 /usr/libexec/gsd-xsettings
root         1849  0.0  0.9 1983104 78892 ?       Ssl  Jan09   4:46 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/co
ntainerd.sock
student    833801  0.0  0.7 826372 63652 ?        Sl   Feb24   0:00 /usr/libexec/evolution-data-server/evolution-alarm-notify
root          355  0.0  0.7 112708 59212 ?        S<s  Jan09   0:38 /usr/lib/systemd/systemd-journald
student    834352  0.0  0.7 3068616 56904 ?       Sl   Feb24   1:05 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com
/app/ding.js -E -P /usr/share/gnome-shell/extensions/ding@rastersoft.com/app
netman@netman-VirtualBox:~/Documents/python_code/lab6$
```

```
netman@netman-VirtualBox:~/Documents/python_code/lab6$ cat RealMadrid_output
Kernel IP routing table
Destination     Gateway         Genmask         Flags   MSS Window  irtt Iface
0.0.0.0         10.224.76.1     0.0.0.0         UG        0 0          0 ens33
10.224.76.0     0.0.0.0         255.255.252.0   U         0 0          0 ens33
172.17.0.0      0.0.0.0         255.255.0.0     U         0 0          0 docker0
192.168.122.0   0.0.0.0         255.255.255.0   U         0 0          0 virbr0
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
jenkins      1487  0.1  4.5 4700188 371468 ?      Ssl  Jan09  83:27 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/ja
va/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080
root         2839  0.4  3.9 2004664 316224 ?      Sl   Jan09 318:16 falcon-sensor-bpf
student    907787  0.0  2.8 3874116 228164 ?      Ssl  Feb24   2:13 /usr/bin/gnome-shell
root         1751  0.0  0.9 1983104 78720 ?       Ssl  Jan09   4:53 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/co
ntainerd.sock
root          352  0.0  0.7 116840 61732 ?        S<s  Jan09   0:41 /usr/lib/systemd/systemd-journald
student    907999  0.0  0.7 826364 59400 ?        Sl   Feb24   0:00 /usr/libexec/evolution-data-server/evolution-alarm-notify
student    908791  0.0  0.6 699668 50352 ?        Ssl  Feb24   0:02 /usr/libexec/gnome-terminal-server
student    908525  0.0  0.6 3134048 48888 ?       Sl   Feb24   1:06 gjs /usr/share/gnome-shell/extensions/ding@rastersoft.com
/app/ding.js -E -P /usr/share/gnome-shell/extensions/ding@rastersoft.com/app
student    908280  0.0  0.6 275216 48708 ?        S    Feb24   0:00 /usr/bin/Xwayland :0 -rootless -noreset -accessx -core -a
uth /run/user/1001/.mutter-Xwaylandauth.7MW4K3 -listenfd 4 -listenfd 5 -displayfd 6 -initfd 7 -byteswappedclients
student    908586  0.0  0.5 737812 46700 ?        Sl   Feb24   0:00 /usr/libexec/mutter-x11-frames
```

2. Provide screenshots of the relevant portion of the playbook and successful file transfer. **[5 Points]**

```
- name: My first play
  hosts: clubhosts
  vars_prompt:
    - name: ssh_password
      prompt: "Enter the ssh password for scp"
```

```
    - name: get routing table
      become: true
      ansible.builtin.shell:
        cmd: "netstat -rn > /{{ inventory_hostname }}_Admin/Route_Info/shell_output.txt"

    - name: get top talkers
      become: true
      ansible.builtin.shell:
        cmd: "ps aux --sort=-%mem | head -n 11 >> /{{ inventory_hostname }}_Admin/Route_Info/shell_output.txt"

    - name: scp output file
      ansible.builtin.expect:
        command: "sudo scp /{{ inventory_hostname }}_Admin/Route_Info/shell_output.txt netman@10.224.77.6:/home/netman/Docum
ents/python_code/lab6/{{ inventory_hostname }}_output"
        responses:
          "netman@10.224.77.6's password":
            - "{{ ssh_password }}"
                                                                                                  114,13        Bot
```

Objective 3.5 – Combining multiple files into a single file

1. Manually generate SSH keys for the following players on their local machines: **[5 Points]**
   - **a. <u>Liverpool</u>**
     - Michael Owen
     - Daniel Agger
     - Steven Gerrard
   - **b. <u>Real Madrid</u>**
     - Luis Figo
     - Fabio Cannavaro
     - Zinedine Zidane

```
student@csci5180-vm3-zaco6003:~$ sudo -u dagger ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/dagger/.ssh/id_ed25519):
Created directory '/home/dagger/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/dagger/.ssh/id_ed25519
Your public key has been saved in /home/dagger/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:QNAMD8x2lT6I4Z03oO0IcCkjtJrZgTJuJTFsp8iwrJ0 dagger@csci5180-vm3-zaco6003
The key's randomart image is:
+--[ED25519 256]--+
|o+ ++=....       |
|*+*.=+= .        |
|@O++ Bo=         |
|*O+.+ =.=        |
|=+.o o .So       |
|o E . .          |
|                 |
|                 |
|                 |
+----[SHA256]-----+
student@csci5180-vm3-zaco6003:~$ sudo -u mowen ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/mowen/.ssh/id_ed25519):
Created directory '/home/mowen/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/mowen/.ssh/id_ed25519
Your public key has been saved in /home/mowen/.ssh/id_ed25519.pub
The key fingerprint is:
```

```
student@csci5180-vm2-zaco6003:~$ sudo -u lfigo ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/lfigo/.ssh/id_ed25519):
Created directory '/home/lfigo/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/lfigo/.ssh/id_ed25519
Your public key has been saved in /home/lfigo/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:KrYjJhrE+MJOa06y/AWDBb7bSg3vl2NAsY6DrU5I+Yk lfigo@csci5180-vm2-zaco6003
The key's randomart image is:
+--[ED25519 256]--+
| .               |
|. ..             |
| . .o            |
|o =o             |
|oX+o    S        |
|*+Xo+  .         |
|=Eo*+.o          |
|@=*.oB           |
|BX.++..          |
+----[SHA256]-----+
student@csci5180-vm2-zaco6003:~$
student@csci5180-vm2-zaco6003:~$ sudo -u fcannavaro ssh-keygen
Generating public/private ed25519 key pair.
Enter file in which to save the key (/home/fcannavaro/.ssh/id_ed25519):
Created directory '/home/fcannavaro/.ssh'.
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/fcannavaro/.ssh/id_ed25519
Your public key has been saved in /home/fcannavaro/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:TB1Jx4cXQNpXGynMzxeiZr3aXXDOCzmNizpzuI6dErk fcannavaro@csci5180-vm2-zaco6003
The key's randomart image is:
```

2. In the same playbook, use the assemble module of Ansible to combine the public keys(.pub) for the above players into a single file on the host machine named as TeamName_authorizedKeys.txt and store the file in the .ssh directory of the root user. (eg. Liverpool_authorizedKeys.txt will have public keys of the three Liverpool players and will be stored in the root user's .ssh directory on the Liverpool machine.)
   [10 Points]
   https://stackoverflow.com/questions/76222429/how-do-i-make-ansible-assemble-output-the-assembled-file-only-on-my-local-machin

```
TASK [create key directory] ******************
ok: [RealMadrid]
ok: [Liverpool]

TASK [copy .ssh files into one directroy] ****
changed: [Liverpool] => (item=mowen)
changed: [RealMadrid] => (item=lfigo)
changed: [Liverpool] => (item=dagger)
changed: [RealMadrid] => (item=fcannavaro)
changed: [Liverpool] => (item=sgerrard)
changed: [RealMadrid] => (item=zzidane)

TASK [assemble files] ************************
changed: [RealMadrid]
changed: [Liverpool]
```

```
student@csci5180-vm2-zaco6003:~$ sudo ls /root/.ssh/
authorized_keys  known_hosts  known_hosts.old  RealMadrid_authorizedKeys.txt
student@csci5180-vm2-zaco6003:~$ sudo cat /root/.ssh/RealMadrid_authorizedKeys.txt
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIIcmD31vpDSyNVj4ORYDRcOE/pTEgcqHK7SG4342nd05 fcannavaro@csci5180-vm2-zaco6003
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIIsQRbJmD4oVyY9esQoFR1vqHRyTTEycZd08dy4jjxWx lfigo@csci5180-vm2-zaco6003
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIHIKumu+ailbcociWMy6FDIOZohTQIqu8IC0j+LAaHWL zzidane@csci5180-vm2-zaco6003
```

```
student@csci5180-vm3-zaco6003:~$ sudo ls /root/.ssh/
authorized_keys  id_ed25519  id_ed25519.pub  known_hosts  known_hosts.old  Liverpool_authorizedKeys.txt
student@csci5180-vm3-zaco6003:~$ sudo cat /root/.ssh/Liverpool_authorizedKeys.txt
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIMyOkhhwfeb2cmqGIksKxTwOgcK44AZnrOxji4n78ksV dagger@csci5180-vm3-zaco6003
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIDiWjwGgDCwaEEN8Ocf4v8OQxP4hyqDEC5SSPmDdpUV0 mowen@csci5180-vm3-zaco6003
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBnrF4ZoTPHcvYhfpokFnf3as16Uh7DxiREsvFhQC3pJ sgerrard@csci5180-vm3-zaco6003
```

3. Using the playbook, send these files to the FIFA_admin machine and store them in the root user's .ssh directory. **[10 Points]**
   https://www.middlewareinventory.com/blog/run-ansible-playbook-locally/
   https://labex.io/tutorials/ansible-how-to-effectively-use-the-local-action-module-in-ansible-414868

```
TASK [copy .ssh files into one directroy] ******************************************************************
changed: [RealMadrid] => (item=lfigo)
changed: [Liverpool] => (item=mowen)
changed: [RealMadrid] => (item=fcannavaro)
changed: [Liverpool] => (item=dagger)
changed: [Liverpool] => (item=sgerrard)
changed: [RealMadrid] => (item=zzidane)

TASK [assemble files] ******************************************************************************************
ok: [Liverpool]
ok: [RealMadrid]

TASK [scp ssh file] ********************************************************************************************
changed: [Liverpool]
changed: [RealMadrid]

PLAY [copy files to root] **************************************************************************************

TASK [Gathering Facts] *****************************************************************************************
ok: [127.0.0.1]

TASK [move files] **********************************************************************************************
changed: [127.0.0.1]

PLAY RECAP *****************************************************************************************************
127.0.0.1                  : ok=2    changed=1    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
Liverpool                  : ok=5    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
RealMadrid                 : ok=5    changed=2    unreachable=0    failed=0    skipped=0    rescued=0    ignored=0
```

4. Provide screenshots of the relevant portion of the playbook and the files in the .ssh directory of the root user of FIFA_admin machine. **[5 Points]**

```
netman@netman-VirtualBox:~/Documents/python_code/lab6$ sudo ls /root/.ssh
id_rsa  id_rsa.pub  Liverpool_authorizedKeys.txt  RealMadrid_authorizedKeys.txt
```

```yaml
  - name: copy .ssh files into one directroy
    ansible.builtin.shell:
      cmd: "cp /home/{{ item }}/.ssh/id_ed25519.pub /tmp/keys/{{ inventory_hostname }}_{{ item }}_authorizedKeys.txt"
    become: true
    loop:
      - "{{ attacker2 }}"
      - "{{ defender2 }}"
      - "{{ captain }}"
    tags:
      - objective3.5

  - name: assemble files
    become: true
    ansible.builtin.assemble:
      dest: "/root/.ssh/{{ inventory_hostname }}_authorizedKeys.txt"
      src: "/tmp/keys"
    tags:
      - objective3.5

  - name: scp ssh file
    become: true
    ansible.builtin.expect:
      command: "sudo scp /root/.ssh/{{ inventory_hostname }}_authorizedKeys.txt netman@10.224.77.6:/tmp/keys/"
      responses:
        "netman@10.224.77.6's password":
          - "{{ ssh_password }}"
    tags:
      - objective3.5

name: copy files to root
hosts: 127.0.0.1
tasks:
  - name: move files
    ansible.builtin.shell:
      cmd: "cp /tmp/keys/* /root/.ssh/"
    become: true
    tags:
      - objective3.5
```

Objective 4 – Extra Credit - Ansible Additional Server Configuration and Router Configurations

Part 1 (Using additional modules in Ansible)

1. Create two more playbooks named as LiverpoolManager.yml and RealMadridManager.yml

2. Think of a use case that will make use of the following Ansible modules in the respective YAML files: **[10 Points]**

File	Modules to use
LiverpoolManager.yml	group_by, wait_for, set_fact
RealMadridManager.yml	template

3. Once you are done executing the above YAML files separately, think of executing all the above tasks via a single playbook that you created for this lab (YourIdentikey_play.yml) **[5 Points]**

Part 2 (Router configuration via Ansible)

1. Construct the same topology done in Automation 1 and configure IBGP between the routers by passing commands through Ansible playbook. It is not required to use multithreading in this case. **[15 Points]**

Total Points _____ / 175